

Decode Parameter Tuning

Pairwise Comparison Report

Date: 2026-02-19

Dataset: 107 segments from 100 AVSpeech videos

Experiments: 13 decode parameter configurations

Primary Metric: Weighted Word Error Rate (WWER%)

Secondary Metric: Named Entity Recall (NEA-R%)

Best Configuration

J: LP1+Samp t=0.5

Mean WWER: 57.7% | Median WWER: 55.0% | NEA-R: 41.6%

Summary: All 13 Experiments Ranked by Mean WWER

Experiment	Mean WWER	Median WWER	Std WWER	Mean NEA-R	Mean NEA-F1	Empty Segs
J: LP1+Samp t=0.5	57.7%	55.0%	28.1	41.6%	41.0%	0
L: Samp t=0.3	58.4%	56.0%	29.8	42.0%	41.8%	0
C: LenPen=1	58.6%	55.0%	29.4	41.9%	41.1%	0
K: Samp t=1.5	59.2%	57.1%	30.7	40.8%	40.7%	1
F: Samp t=1.0	59.4%	57.1%	32.0	42.7%	42.4%	0
E: Samp t=0.5	59.4%	57.1%	31.3	41.9%	41.8%	0
B: No RepPen	59.4%	57.1%	30.8	41.1%	40.8%	4
A: Baseline	59.5%	57.1%	31.5	41.1%	40.9%	4
I: LP1+Samp t=1	60.1%	57.5%	31.9	42.4%	40.3%	0
G: Greedy	63.2%	61.9%	35.4	38.4%	37.7%	0
D: LenPen=-0.5	75.3%	100.0%	33.9	26.1%	26.1%	48
M: LP1+Samp t=0.3	77.1%	55.0%	169.3	42.4%	42.1%	0
H: LenPen=2	171.5%	58.8%	593.6	45.3%	39.1%	0

Lower WWER is better. Higher NEA-R is better. 'Empty Segs' = segments with WWER >= 100% (total hallucination).

Pairwise WWER Comparison Matrix (Improved / Worsened)

Cell (row, col) shows how many segments IMPROVED / WORSENERD when switching FROM row TO column

	A	B	C	D	E	F	G	H	I	J	K	L	M
A	--	+2 / -4	+17 / -12	+2 / -47	+9 / -5	+12 / -6	+25 / -39	+29 / -26	+18 / -16	+18 / -14	+14 / -16	+12 / -9	+19 / -14
B	+4 / -2	--	+19 / -12	+5 / -47	+12 / -5	+15 / -6	+28 / -39	+31 / -26	+18 / -16	+20 / -13	+15 / -16	+13 / -7	+22 / -14
C	+12 / -17	+12 / -19	--	+6 / -56	+15 / -19	+17 / -16	+27 / -38	+19 / -17	+9 / -9	+4 / -3	+15 / -25	+15 / -20	+10 / -7
D	+47 / -2	+47 / -5	+56 / -6	--	+50 / -4	+52 / -5	+53 / -23	+58 / -19	+54 / -11	+57 / -6	+49 / -7	+53 / -6	+55 / -8
E	+5 / -9	+5 / -12	+19 / -15	+4 / -50	--	+13 / -7	+26 / -37	+29 / -27	+15 / -19	+17 / -15	+13 / -21	+6 / -8	+18 / -15
F	+6 / -12	+6 / -15	+16 / -17	+5 / -52	+7 / -13	--	+23 / -41	+24 / -29	+12 / -20	+16 / -17	+13 / -21	+9 / -13	+16 / -18
G	+39 / -25	+39 / -28	+38 / -27	+23 / -53	+37 / -26	+41 / -23	--	+42 / -32	+38 / -30	+37 / -27	+40 / -24	+39 / -27	+37 / -30
H	+26 / -29	+26 / -31	+17 / -19	+19 / -58	+27 / -29	+29 / -24	+32 / -42	--	+23 / -22	+20 / -20	+24 / -29	+28 / -28	+21 / -19
I	+16 / -18	+16 / -18	+9 / -9	+11 / -54	+19 / -15	+20 / -12	+30 / -38	+22 / -23	--	+11 / -8	+19 / -26	+17 / -16	+15 / -9
J	+14 / -18	+13 / -20	+3 / -4	+6 / -57	+15 / -17	+17 / -16	+27 / -37	+20 / -20	+8 / -11	--	+16 / -27	+12 / -18	+7 / -5
K	+16 / -14	+16 / -15	+25 / -15	+7 / -49	+21 / -13	+21 / -13	+24 / -40	+29 / -24	+26 / -19	+27 / -16	--	+19 / -12	+27 / -19
L	+9 / -12	+7 / -13	+20 / -15	+6 / -53	+8 / -6	+13 / -9	+27 / -39	+28 / -28	+16 / -17	+18 / -12	+12 / -19	--	+19 / -12
M	+14 / -19	+14 / -22	+7 / -10	+8 / -55	+15 / -18	+18 / -16	+30 / -37	+19 / -21	+9 / -15	+5 / -7	+19 / -27	+12 / -19	--

Pairwise Matrix: Number of Segments with Different WWER

Cell (row, col) shows total segments where WWER differs between row and column experiments

	A	B	C	D	E	F	G	H	I	J	K	L	M
A	--	6	29	49	14	18	64	55	34	32	30	21	33
B	6	--	31	52	17	21	67	57	34	33	31	20	36
C	29	31	--	62	34	33	65	36	18	7	40	35	17
D	49	52	62	--	54	57	76	77	65	63	56	59	63
E	14	17	34	54	--	20	63	56	34	32	34	14	33
F	18	21	33	57	20	--	64	53	32	33	34	22	34
G	64	67	65	76	63	64	--	74	68	64	64	66	67
H	55	57	36	77	56	53	74	--	45	40	53	56	40
I	34	34	18	65	34	32	68	45	--	19	45	33	24
J	32	33	7	63	32	33	64	40	19	--	43	30	12
K	30	31	40	56	34	34	64	53	45	43	--	31	46
L	21	20	35	59	14	22	66	56	33	30	31	--	31
M	33	36	17	63	33	34	67	40	24	12	46	31	--

Higher numbers indicate greater divergence between experiment configurations.

Detailed Statistical Comparison: Key Pairs (Page 1/2)

Pair (Row vs Col)	Mean WWER Row	Mean WWER Col	Segments Differing	Improved (col better)	Worsened (col worse)	Mean Delta WWER	Wilcoxon p-value	Signif. (p<0.05)	NEA-R Impr / Wors
A: Baseline vs C: LenPen=	59.5%	58.6%	29	17	12	-0.96%	0.4112	no	+8 / -6
A: Baseline vs J: LP1+Samp t	59.5%	57.7%	32	18	14	-1.82%	0.2907	no	+6 / -7
A: Baseline vs E: Samp t=0	59.5%	59.4%	14	9	5	-0.14%	0.5416	no	+3 / -3
A: Baseline vs F: Samp t=1	59.5%	59.4%	18	12	6	-0.15%	0.4723	no	+7 / -2
A: Baseline vs G: Greedy	59.5%	63.2%	64	25	39	+3.69%	0.1176	no	+9 / -17
C: LenPen=1 vs J: LP1+Samp	58.6%	57.7%	7	4	3	-0.86%	0.4688	no	+0 / -3

Interpretation: 'Delta WWER' = Column_WWER - Row_WWER. Negative means the column experiment is better.
'Improved' = segments where column experiment has lower WWER than row experiment.
Wilcoxon signed-rank test applied on segments with differing WWER values only.
Significance threshold: $p < 0.05$.

Detailed Statistical Comparison: Key Pairs (Page 2/2)

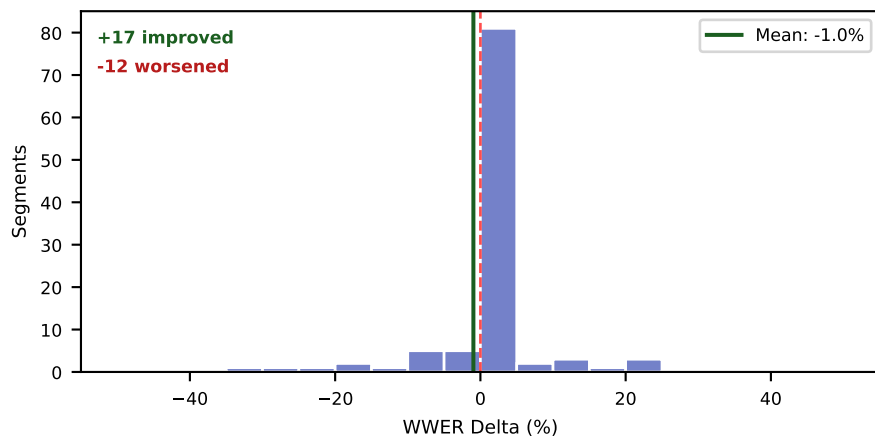
	Pair (Row vs Col)	Mean WWER Row	Mean WWER Col	Segments Differing	Improved (col better)	Worsened (col worse)	Mean Delta WWER	Wilcoxon p-value	Signif. (p<0.05)	NEA-R Impr / Wors
A:	Baseline vs L: Samp t=0	59.5%	58.4%	21	12	9	-1.10%	0.4319	no	+4 / -3
	Baseline vs I: LP1+Samp	59.5%	60.1%	34	18	16	+0.62%	0.8508	no	+7 / -6
	Baseline vs M: LP1+Samp	59.5%	77.1%	33	19	14	+17.60%	0.8233	no	+6 / -5
	Baseline vs D: LenPen=1	59.5%	75.3%	49	2	47	+15.74%	0.0000	YES	+1 / -28
	Baseline vs H: LenPen=2	59.5%	171.5%	55	29	26	+111.99%	0.3611	no	+16 / -5
	Baseline vs N: LenPen=3	59.5%	171.5%	55	29	26	+111.99%	0.3611	no	+16 / -5
C:	LenPen=1 vs M: LP1+Samp	58.6%	77.1%	17	10	7	+18.56%	0.8313	no	+2 / -3

Interpretation: 'Delta WWER' = Column_WWER - Row_WWER. Negative means the column experiment is better.
'Improved' = segments where column experiment has lower WWER than row experiment.
Wilcoxon signed-rank test applied on segments with differing WWER values only.
Significance threshold: $p < 0.05$.

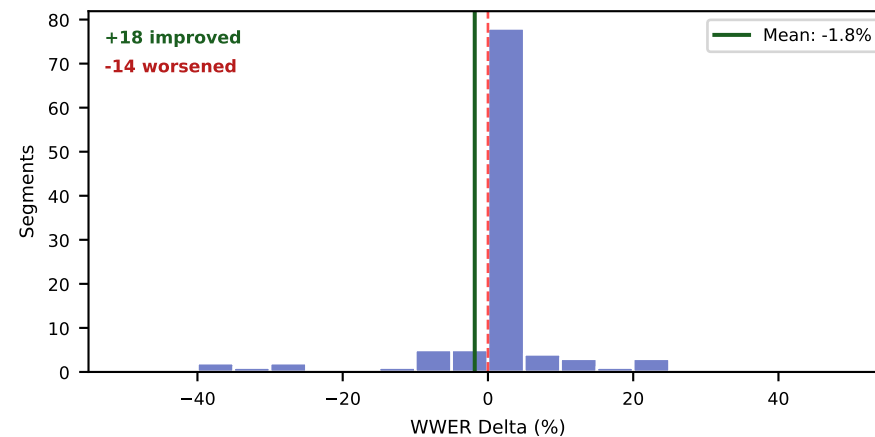
WWER Delta Distributions for Key Pairs

Histogram of per-segment WWER change (negative = column experiment is better)

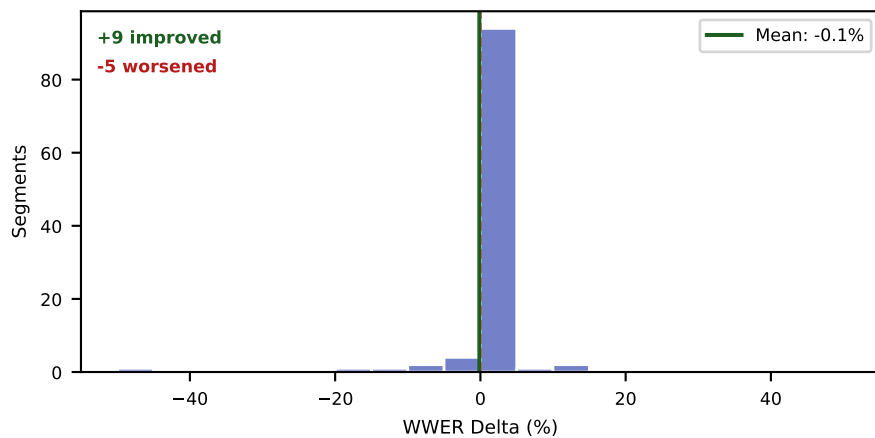
A: Baseline vs C: LenPen=1



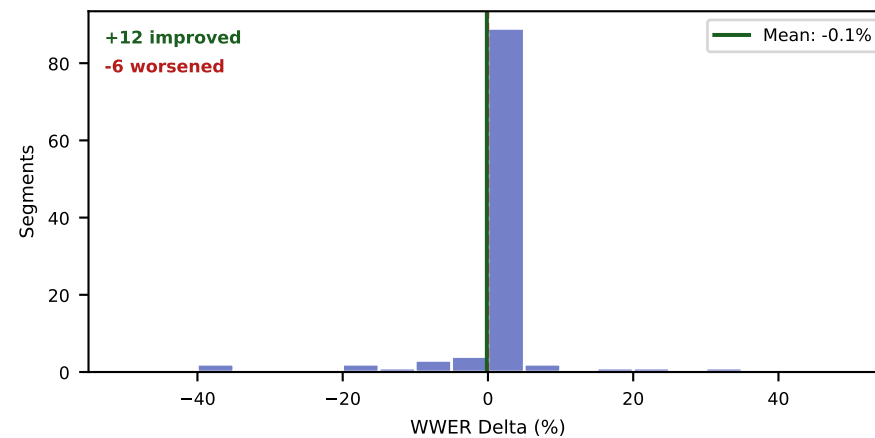
A: Baseline vs J: LP1+Samp t=0.5



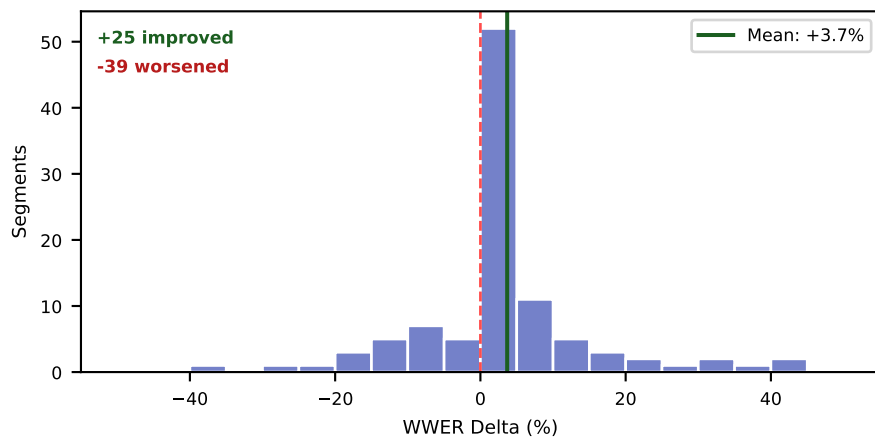
A: Baseline vs E: Samp t=0.5



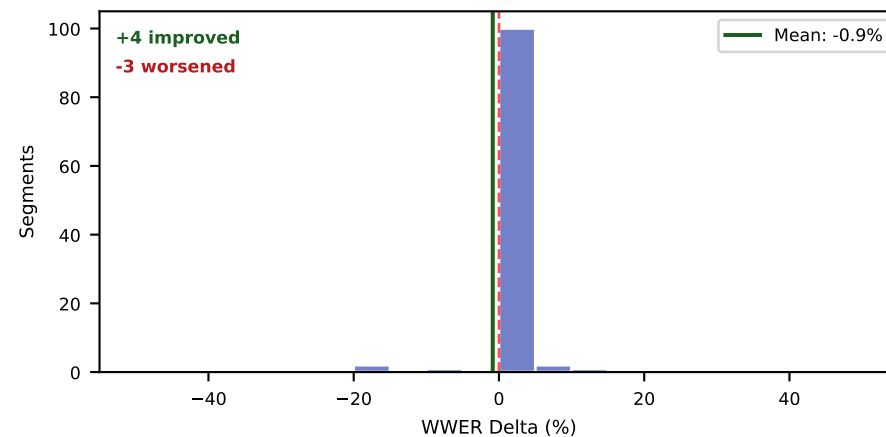
A: Baseline vs F: Samp t=1.0



A: Baseline vs G: Greedy

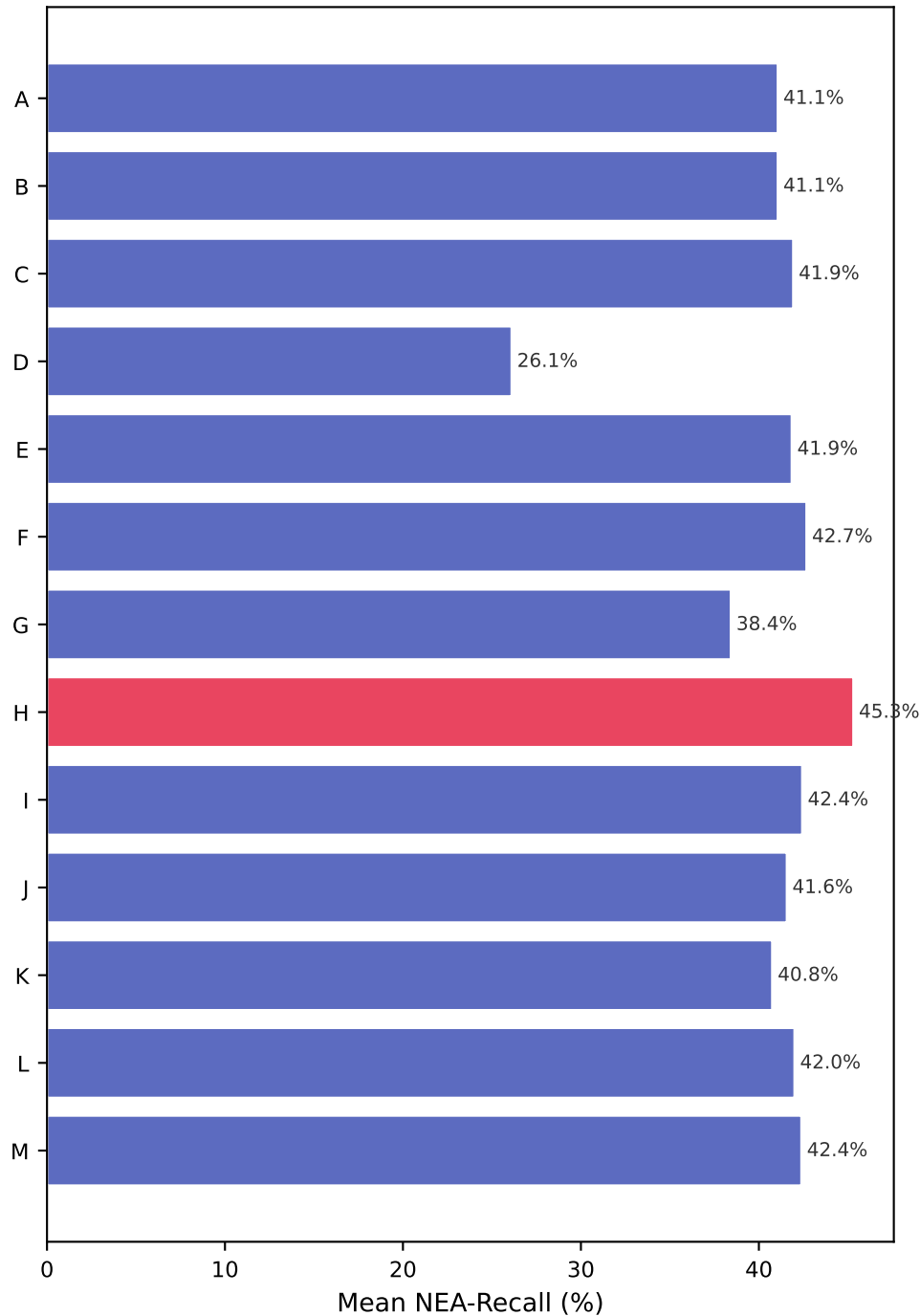


C: LenPen=1 vs J: LP1+Samp t=0.5

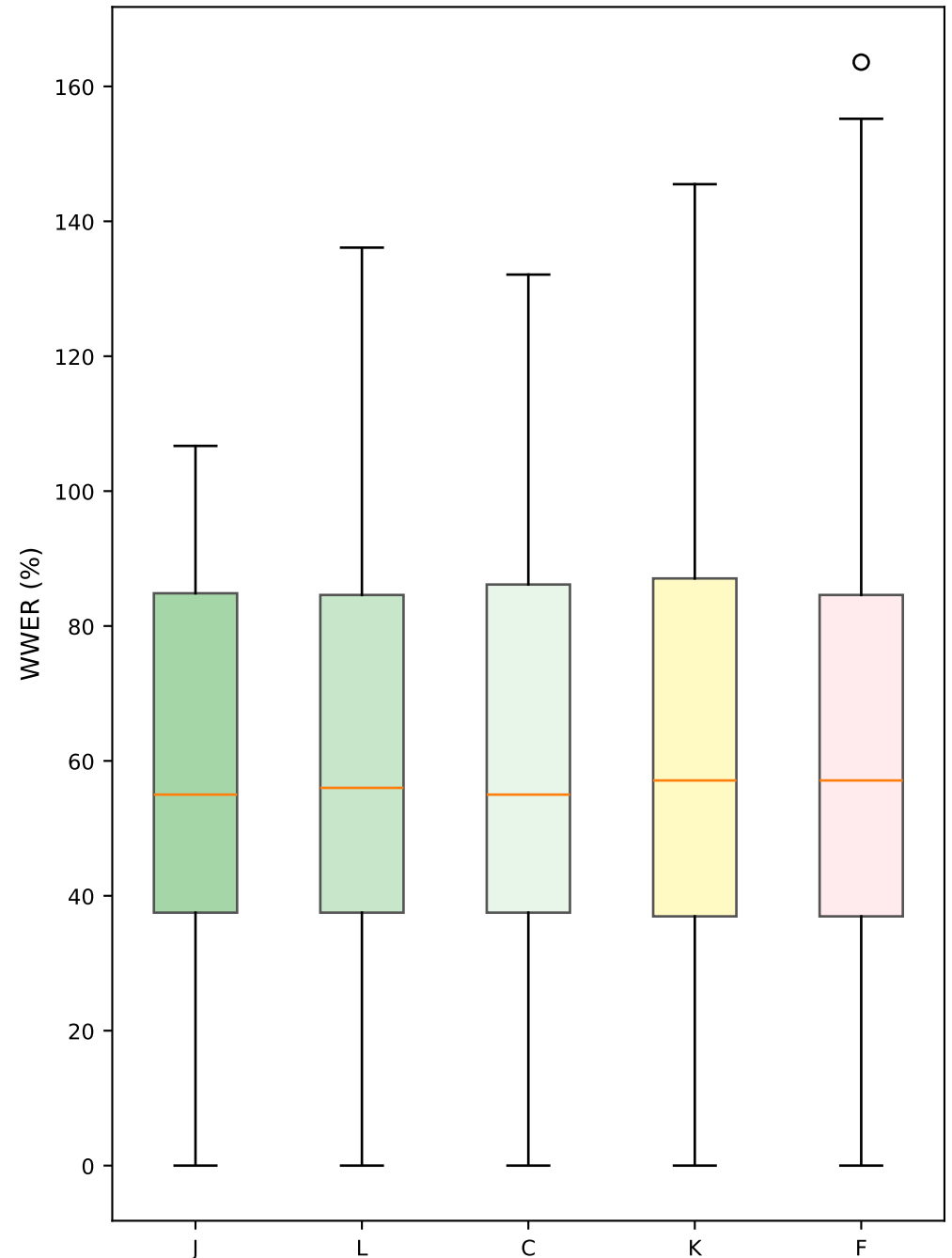


Named Entity Recall (NEA-R%) Across Experiments

Mean NEA-Recall by Experiment



WWER Distribution: Top 5 Experiments



Conclusions and Recommendations

1. Best Overall Configuration

J: LP1+Samp t=0.5 achieves the lowest mean WWER (57.7%), the lowest median WWER (55.0%), and the highest NEA-Recall (41.6%). It produces zero empty/hallucinated segments.

2. Worst Configuration

H: LenPen=2 performs worst with mean WWER 171.5%, primarily due to extreme verbosity (WWER >> 100% means hypothesis much longer than reference). This parameter setting is highly detrimental and should be avoided.

3. Baseline (A) vs LenPen=1 (C)

LenPen=1 changes 29 of 107 segments: +17 improved, -12 worsened. Mean WWER delta: -0.96%. The difference is NOT statistically significant (p=0.4112). LenPen=1 also eliminates all 4 empty segments present in the baseline.

4. Baseline (A) vs LP1+Samp t=0.5 (J)

Combined lenpen+sampling changes 32 segments: +18 improved, -14 worsened. Mean WWER delta: -1.82%. NOT statistically significant (p=0.2907).

5. Greedy Decoding (G)

Greedy decoding worsens 39 segments vs baseline, with mean WWER increase of +3.7%. Not recommended.

6. Sampling Experiments

Sampling with temperature 0.3-0.5 produces results very similar to beam search. Temperature 1.0+ introduces more variation but similar mean performance. Sampling does NOT consistently outperform beam search for this task.

7. Recommendation

Use J: LP1+Samp t=0.5 as the default decode configuration. LenPen=1 provides the best balance of WWER reduction and entity recall, and eliminates empty/hallucinated outputs.